class14: RNA-Seq analysis mini-project

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```r
library(DESeq2)
```

```
Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs
```

```
The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min


Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'
```

```
The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```r
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q. Complete the code below to remove the troublesome first column from countData

```r
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will rowSums() of countData return and how could you use it in this context?

```
countData = countData[rowSums(countData) > 0, ]
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

```
dds = DESeqDataSetFromMatrix(countData=countData,
                             colData=colData,
                             design=~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```
dds = DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```r
res = results(dds)
```

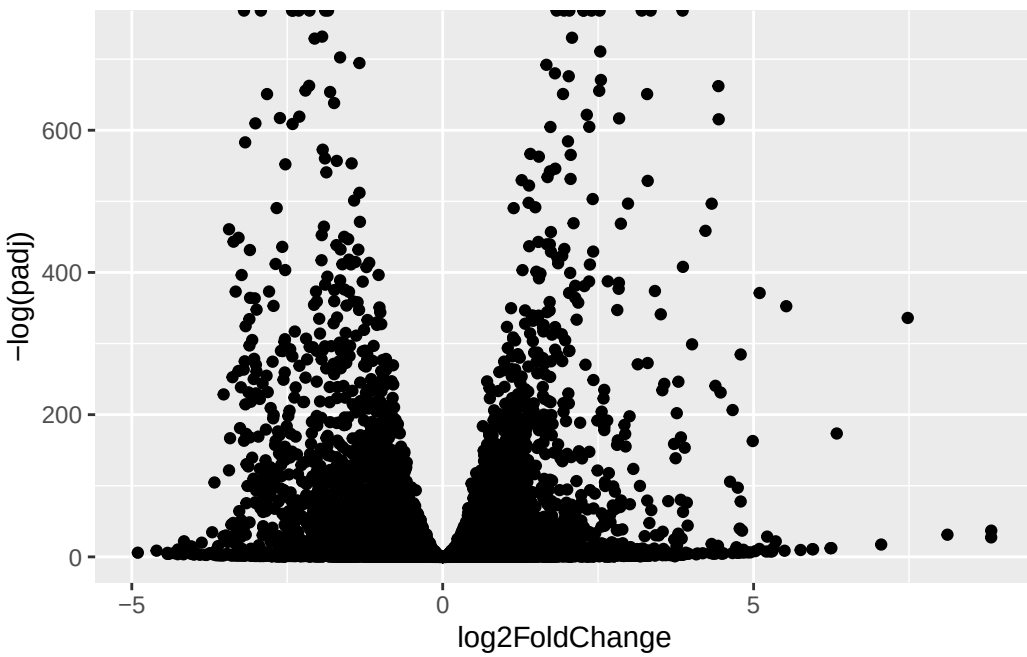
> Q. Call the summary() function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```r
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

```r
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj) ) +
  geom_point()
```

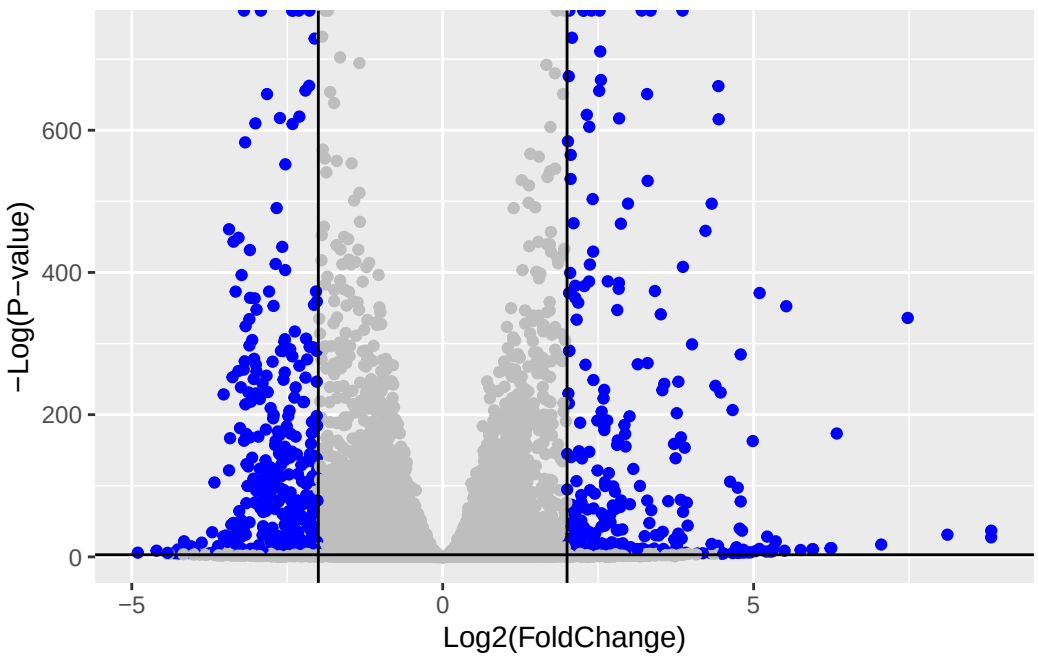
```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Q. Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.05), col = "black")
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol = mapIds(org.Hs.eg.db,
                    keys=rownames(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez = mapIds(org.Hs.eg.db,
                    keys=rownames(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name =   mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype= "ENSEMBL",
                    column= "ENTREZID",
                    multiVals="first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.913579      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.229650      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.188076     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.637938      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.255123      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.979750      0.5428105 0.5215598   1.040744 2.97994e-01
ENSG00000188290  108.922128      2.0570638 0.1969053  10.446970 1.51282e-25
ENSG00000187608  350.716868      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.439422      0.3899088 0.0467163   8.346304 7.04321e-17
ENSG00000237330    0.158192      0.7859552 4.0804729   0.192614 8.47261e-01
                      padj      symbol      entrez        name
                  <numeric> <character> <character> <character>
ENSG00000279457 6.86555e-01          NA          NA          NA
ENSG00000187634 5.15718e-03      SAMD11      148398      148398
ENSG00000188976 1.76549e-35       NOC2L       26155       26155
```

```
ENSG00000187961 1.13413e-07      KLHL17      339451      339451
ENSG00000187583 9.19031e-01     PLEKHN1       84069       84069
ENSG00000187642 4.03379e-01       PERM1       84808       84808
ENSG00000188290 1.30538e-24        HES4       57801       57801
ENSG00000187608 2.37452e-02       ISG15        9636        9636
ENSG00000188157 4.21963e-16        AGRN      375790      375790
ENSG00000237330          NA      RNF223      401934      401934
```

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```r
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")
```

```r
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
library(gage)
```

```r
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

```r
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```r
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```r
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```r
head(keggres$less)
```

```
                                        p.geomean stat.mean        p.val
hsa04110 Cell cycle                   8.995727e-06 -4.378644 8.995727e-06
hsa03030 DNA replication              9.424076e-05 -3.951803 9.424076e-05
hsa03013 RNA transport                1.375901e-03 -3.028500 1.375901e-03
hsa03440 Homologous recombination     3.066756e-03 -2.852899 3.066756e-03
hsa04114 Oocyte meiosis               3.784520e-03 -2.698128 3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis 8.961413e-03 -2.405398 8.961413e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001448312      121 8.995727e-06
hsa03030 DNA replication              0.007586381       36 9.424076e-05
hsa03013 RNA transport                0.073840037      144 1.375901e-03
hsa03440 Homologous recombination     0.121861535       28 3.066756e-03
hsa04114 Oocyte meiosis               0.121861535      102 3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis 0.212222694       53 8.961413e-03
```

```r
head(keggres$greater)
```

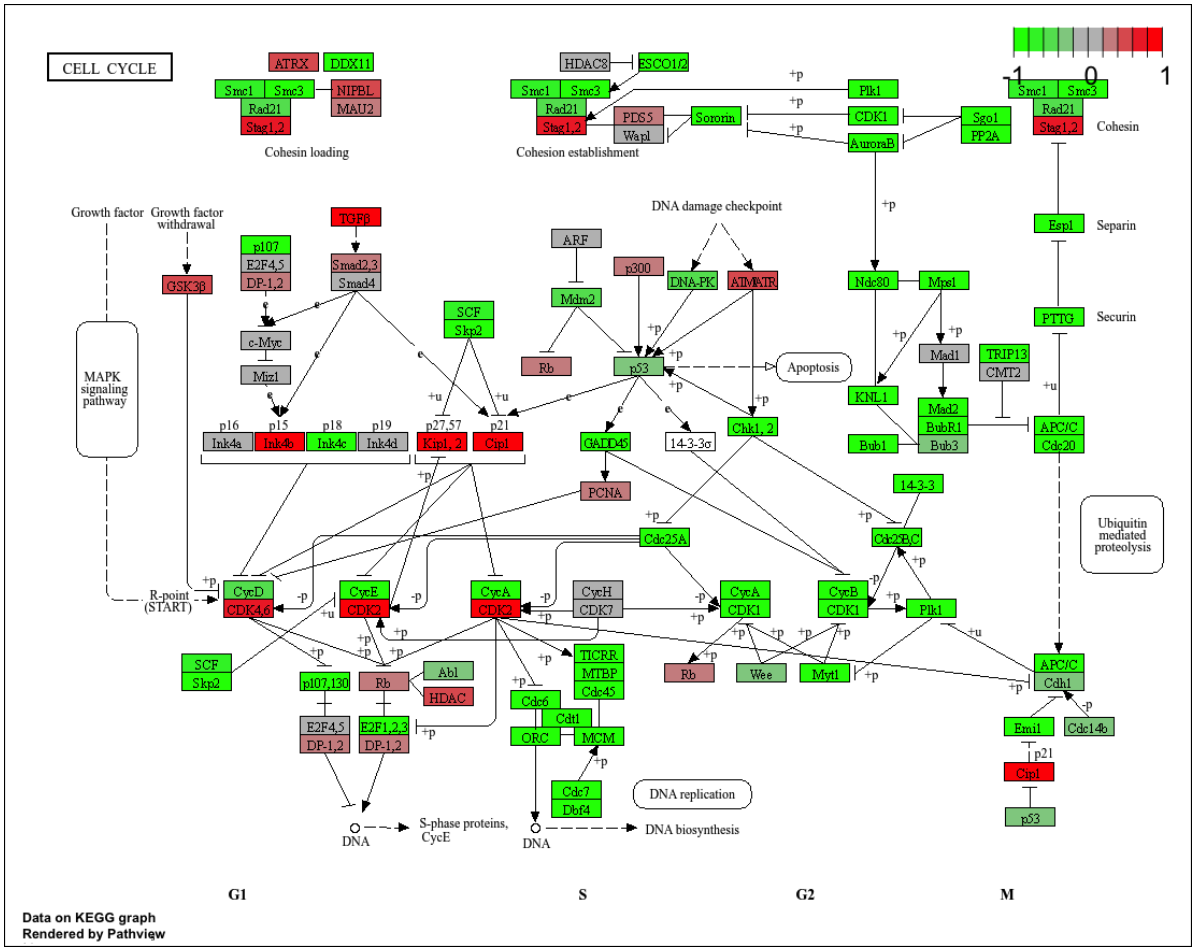
```
                                      p.geomean stat.mean       p.val
hsa04640 Hematopoietic cell lineage   0.002822776  2.833362 0.002822776
hsa04630 Jak-STAT signaling pathway   0.005202070  2.585673 0.005202070
hsa00140 Steroid hormone biosynthesis 0.007255099  2.526744 0.007255099
hsa04142 Lysosome                     0.010107392  2.338364 0.010107392
hsa04330 Notch signaling pathway      0.018747253  2.111725 0.018747253
hsa04916 Melanogenesis                0.019399766  2.081927 0.019399766
                                          q.val set.size        exp1
hsa04640 Hematopoietic cell lineage   0.3893570       55 0.002822776
hsa04630 Jak-STAT signaling pathway   0.3893570      109 0.005202070
hsa00140 Steroid hormone biosynthesis 0.3893570       31 0.007255099
hsa04142 Lysosome                     0.4068225      118 0.010107392
hsa04330 Notch signaling pathway      0.4391731       46 0.018747253
hsa04916 Melanogenesis                0.4391731       90 0.019399766
```

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project
```

```
Info: Writing image file hsa04110.pathview.png
```




```r
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Warning: reconcile groups sharing member nodes!

     [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04110.pathview.pdf
```

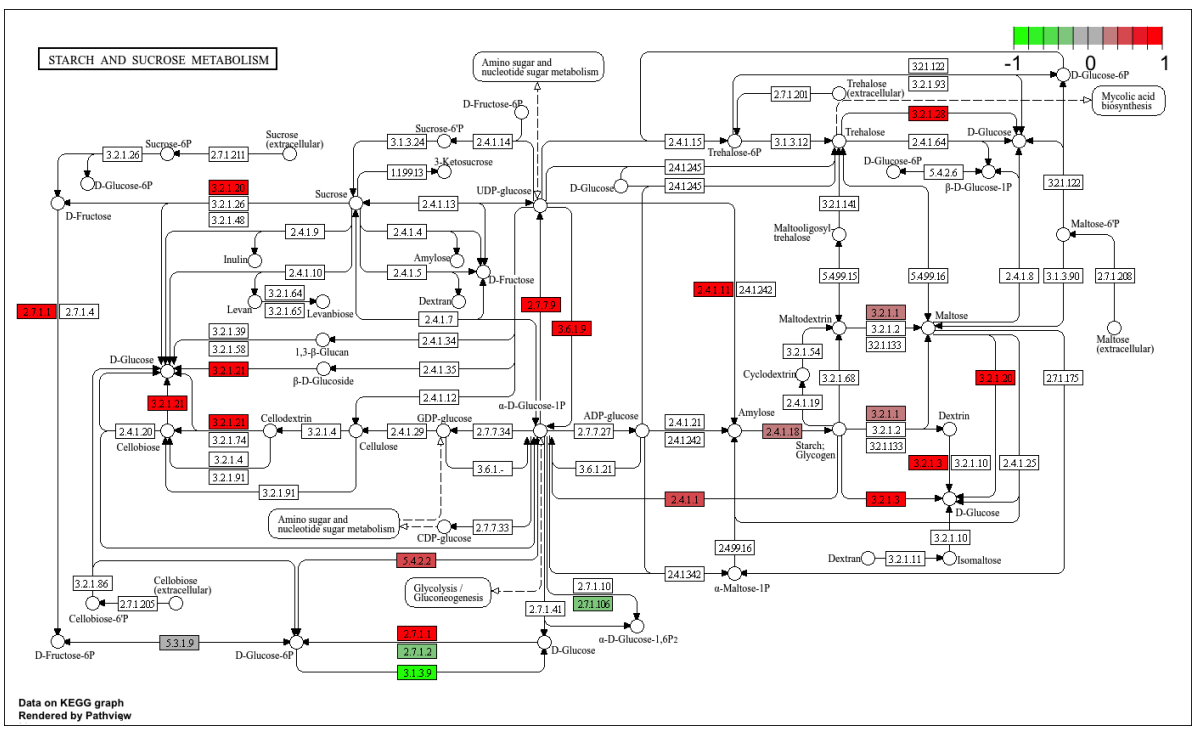
```r
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

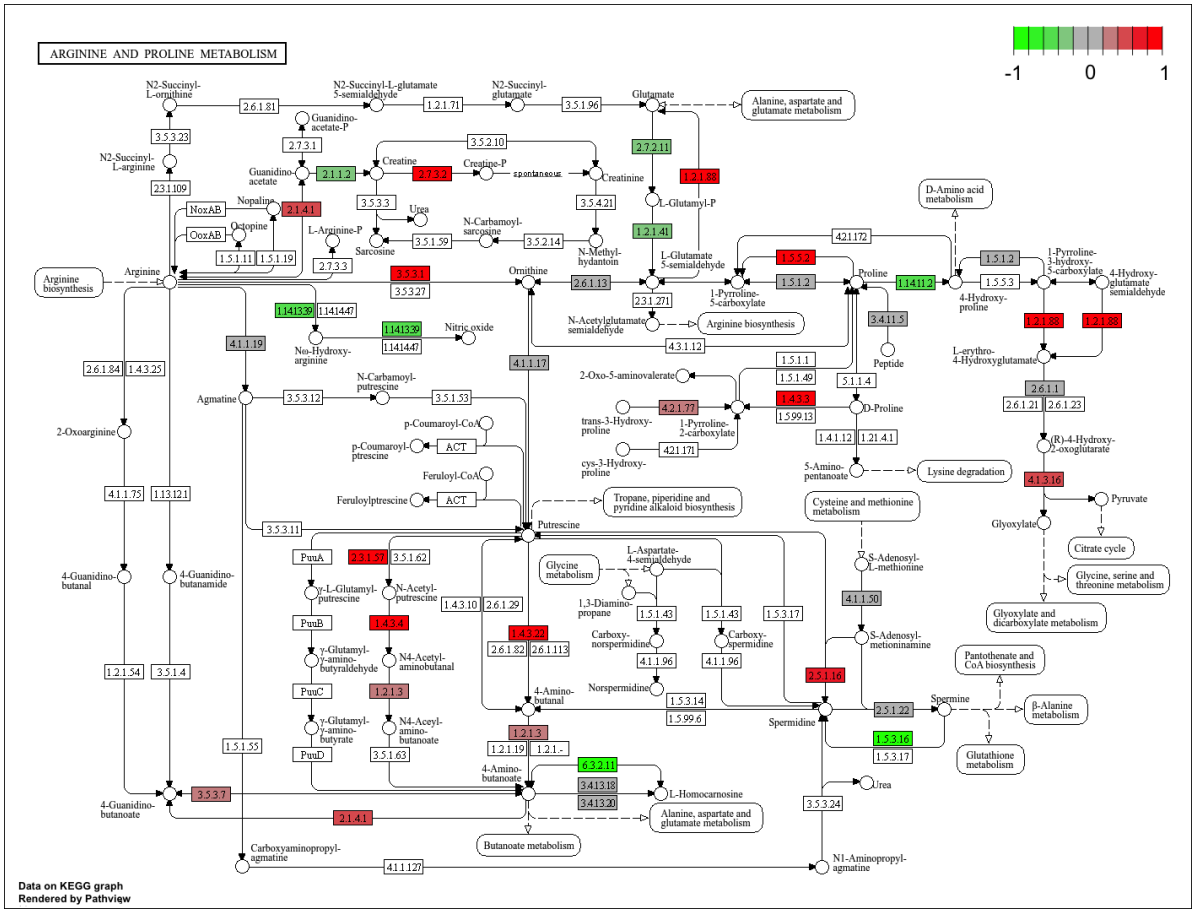
```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

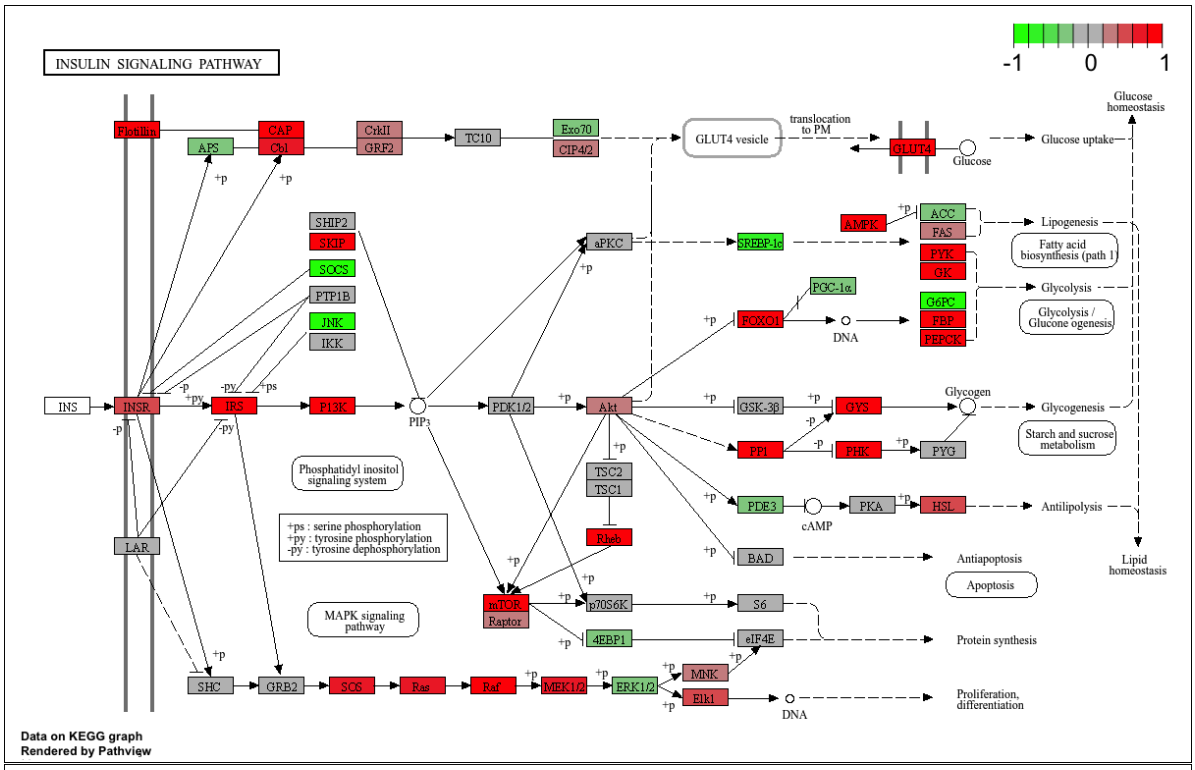
```r
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

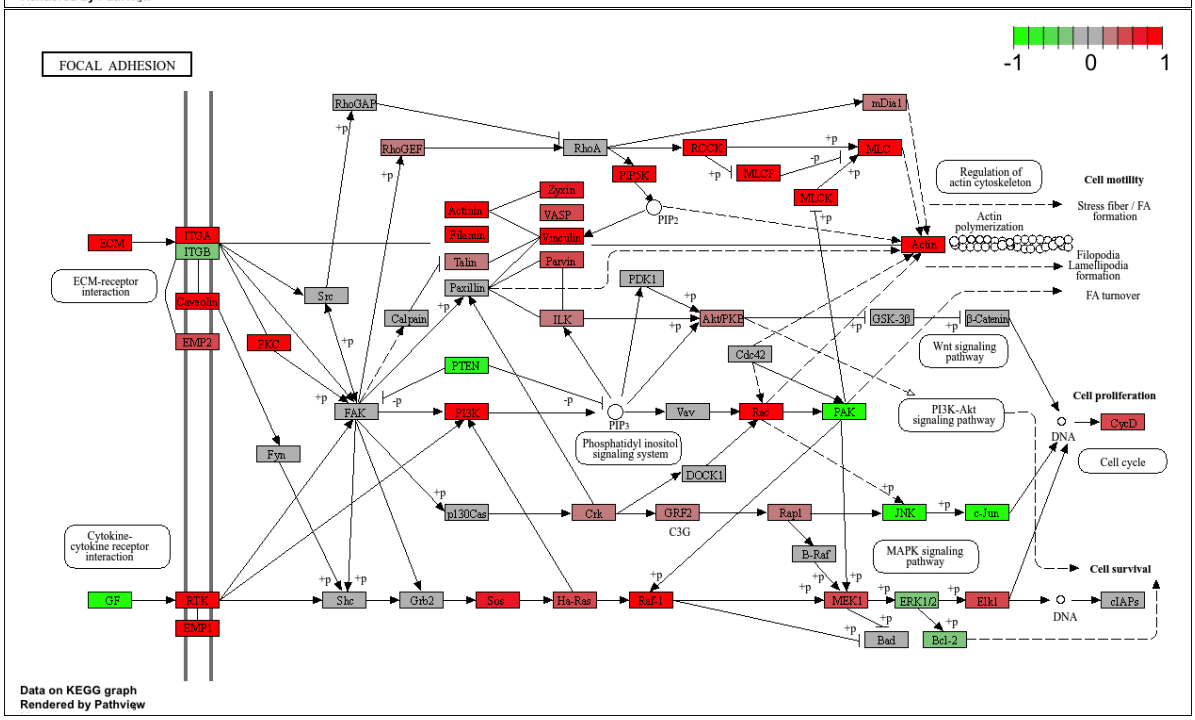
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns
```

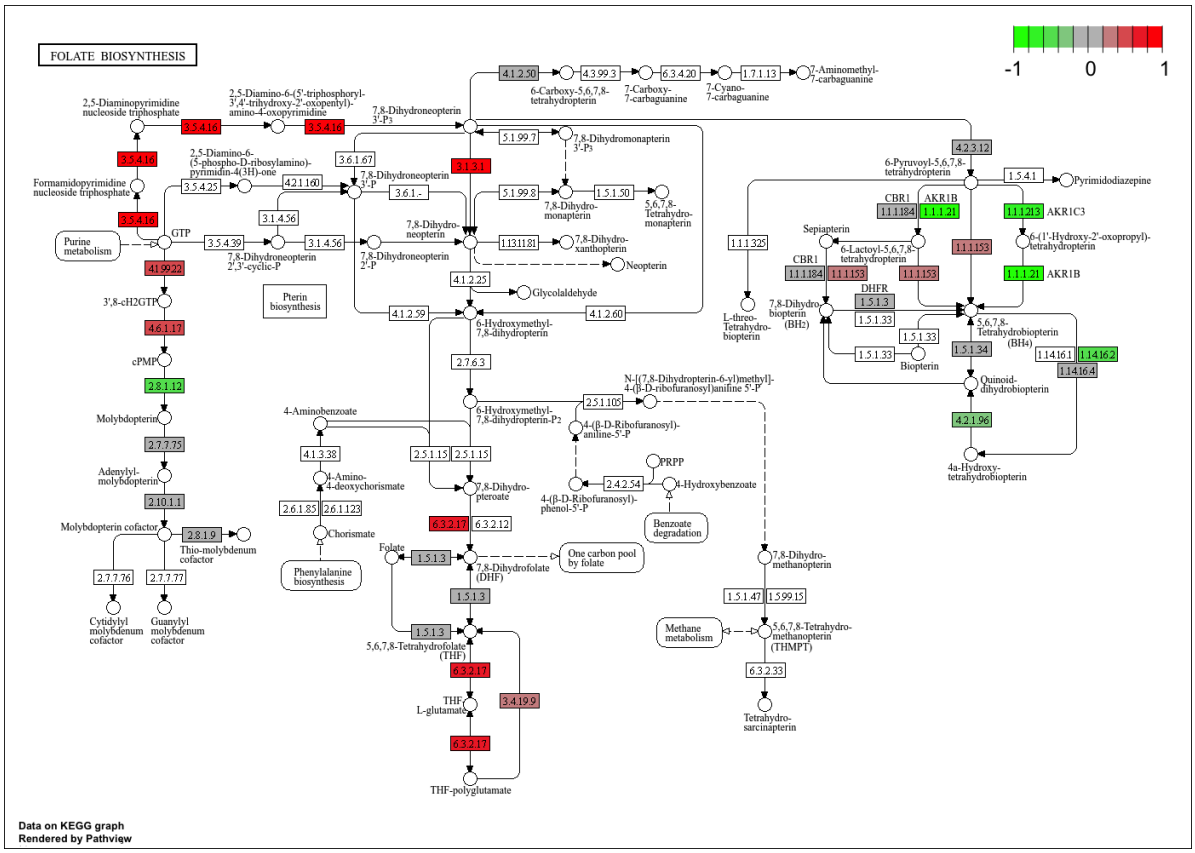
```
Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04330.pathview.png
```




ARGININE AND PROLINE METABOLISM
-1
0
1
N2-Succinyl-L-ornithine
2.6.1.81
N2-Succinyl-L-glutamate 5-semialdehyde
1.2.1.71
N2-Succinyl-glutamate
3.5.1.96
Glutamate
Alanine, aspartate and glutamate metabolism
3.5.3.23
Guanidino-acetate-P
2.7.3.1
3.5.2.10
2.7.2.11
N2-Succinyl-L-arginine
Creatine
Creatine-P
spontaneous
Creatinine
1.2.1.88
Guanidino-acetate
2.1.1.2
2.7.3.2
23.1.109
3.5.3.3
3.5.4.21
L-Glutamyl-P
D-Amino acid metabolism
NoxAB
Nopaline
2.1.4.1
Urea
N-Carbamoyl-sarcosine
1.2.1.41
OoxAB
Octopine
L-Arginine-P
Sarcosine
3.5.1.59
3.5.2.14
N-Methyl-hydantoin
L-Glutamate 5-semialdehyde
4.2.1.172
1.5.5.2
1.5.1.2
1.5.1.11
1.5.1.19
2.7.3.3
Arginine biosynthesis
Arginine
3.5.3.1
Ornithine
2.6.1.13
1-Pyrroline-5-carboxylate
Proline
1.14.11.2
4-Hydroxy-proline
1-Pyrroline-3-hydroxy-5-carboxylate
4-Hydroxy-glutamate semialdehyde
3.5.3.27
2.3.1.271
1.14.13.39
1.14.14.47
N-Acetylglutamate semialdehyde
Arginine biosynthesis
3.4.11.5
1.2.1.88
1.2.1.89
4.1.1.19
Nitric oxide
Neo-Hydroxy-arginine
4.3.1.12
Peptide
L-erythro-4-Hydroxyglutamate
2.6.1.84
1.4.3.25
4.1.1.17
1.5.1.1
1.5.1.49
2-Oxo-5-aminovalerate
5.1.1.4
2.6.1.1
N-Carbamoyl-putrescine
Agmatine
3.5.3.12
3.5.1.53
1.4.3.3
1.5.99.13
D-Proline
2.6.1.21
2.6.1.23
p-Coumaroyl-CoA
trans-3-Hydroxy-proline
4.2.1.77
1-Pyrroline-2-carboxylate
2-Oxoarginine
p-Coumaroyl-ptrescine
ACT
4.2.1.171
1.4.1.12
1.21.4.1
(R)-4-Hydroxy-2-oxoglutarate
Feruloyl-CoA
cys-3-Hydroxy-proline
5-Amino-pentanoate
Lysine degradation
4.1.3.16
4.1.1.75
1.13.12.1
Feruloylptrescine
ACT
Tropane, piperidine and pyridine alkaloid biosynthesis
Cysteine and methionine metabolism
Pyruvate
3.5.3.11
Putrescine
Glyoxylate
Citrate cycle
PuuA
2.3.1.57
3.5.1.62
L-Aspartate-4-semialdehyde
S-Adenosyl-L-methionine
Glycine metabolism
Glycine, serine and threonine metabolism
4-Guanidino-butanal
4-Guanidino-butanamide
γ-L-Glutamyl-putrescine
N-Acetyl-putrescine
1.4.3.10
2.6.1.29
1,3-Diamino-propane
1.5.1.43
1.5.1.43
1.5.3.17
4.1.1.50
Glyoxylate and dicarboxylate metabolism
PuuB
1.4.3.4
1.4.3.22
Carboxy-norspermidine
Carboxy-spermidine
S-Adenosyl-metioninamine
2.6.1.82
2.6.1.113
γ-Glutamyl-γ-amino-butyraldehyde
N4-Acetyl-aminobutanal
4.1.1.96
4.1.1.96
Pantothenate and CoA biosynthesis
1.2.1.54
3.5.1.4
2.5.1.16
PuuC
1.2.1.3
Norspermidine
4-Amino-butanal
1.5.3.14
1.5.99.6
2.5.1.22
Spermine
β-Alanine metabolism
γ-Glutamyl-γ-amino-butyrate
N4-Aceyl-amino-butanoate
Spermidine
1.5.3.16
1.5.3.17
1.5.1.55
1.2.1.3
1.2.1.19
1.2.1.-
PuuD
3.5.1.63
Glutathione metabolism
6.3.2.11
4-Amino-butanoate
3.4.13.18
3.4.13.20
L-Homocarnosine
Urea
4-Guanidino-butanoate
3.5.3.7
3.5.3.24
2.1.4.1
Alanine, aspartate and glutamate metabolism
Butanoate metabolism
Carboxyaminopropyl-agmatine
4.1.1.127
N1-Aminopropyl-agmatine
Data on KEGG graph
Rendered by Pathview


INSULIN SIGNALING PATHWAY
-1
0
1
Flotillin
APS
CAP
Cbl
CrkII
GRF2
TC10
Exo70
CIP4/2
GLUT4 vesicle
translocation to PM
GLUT4
Glucose
Glucose uptake
Glucose homeostasis
SHIP2
SKIP
SOCS
PTP1B
JNK
IKK
aPKC
SREBP-1c
AMPK
ACC
FAS
PYK
GK
Lipogenesis
Fatty acid biosynthesis (path 1)
PGC-1α
FOXO1
DNA
G6PC
FBP
PEPCK
Glycolysis
Glycolysis / Glucone ogenesis
INS
INSR
IRS
PI3K
PIP3
PDK1/2
Akt
GSK-3β
GYS
Glycogen
Glycogenesis
PP1
PHK
PYG
Starch and sucrose metabolism
Phosphatidyl inositol signaling system
TSC2
TSC1
PDE3
cAMP
PKA
HSL
Antilipolysis
+ps : serine phosphorylation
+py : tyrosine phosphorylation
-py : tyrosine dephosphorylation
Rheb
LAR
BAD
Antiapoptosis
Apoptosis
Lipid homeostasis
mTOR
Raptor
p70S6K
S6
MAPK signaling pathway
4EBP1
eIF4E
Protein synthesis
SHC
GRB2
SOS
Ras
Raf
MEK1/2
ERK1/2
MNK
Elk1
DNA
Proliferation, differentiation
Data on KEGG graph
Rendered by Pathview


FOCAL ADHESION
-1
0
1
RhoGAP
RhoGEF
RhoA
mDia1
ROCK
MLC
PIP5K
MLCP
MLCK
Regulation of actin cytoskeleton
Cell motility
Stress fiber / FA formation
Actin polymerization
Actin
Filopodia
Lamellipodia formation
FA turnover
Zyxin
VASP
Vinculin
Paxillin
Actinin
Filamin
Talin
Parvin
PIP2
ECM
ITGA
ITGB
ECM-receptor interaction
Caveolin
EMP2
PKC
Src
Calpain
PDK1
ILK
Akt/PKB
GSK-3β
β-Catenin
Wnt signaling pathway
Cdc42
PTEN
FAK
PI3K
PIP3
Vav
Rac
PAK
PI3K-Akt signaling pathway
Cell proliferation
CycD
DNA
Cell cycle
Fyn
Phosphatidyl inositol signaling system
DOCK1
p130Cas
Crk
GRF2
C3G
Rap1
JNK
c-Jun
Cytokine-cytokine receptor interaction
B-Raf
MAPK signaling pathway
Cell survival
GF
RTK
EMP1
Shc
Grb2
Sos
Ha-Ras
Raf-1
MEK1
ERK1/2
Elk1
cIAPs
Bad
Bcl-2
Data on KEGG graph
Rendered by Pathview



Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
## Focus on top 5 downregulated pathways here for demo purposes only
keggrespathwaysd <- rownames(keggres$less)[1:5]
keggresidsd = substr(keggrespathwaysd, start=1, stop=8)
keggresidsd
```

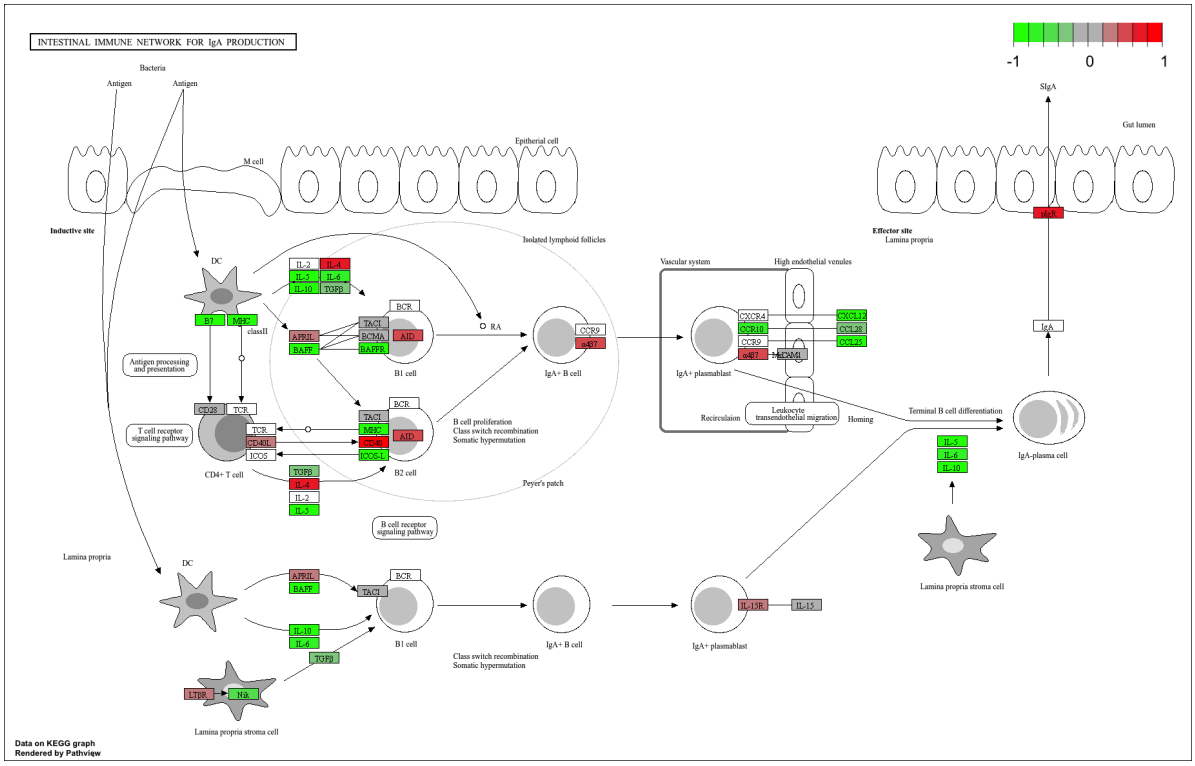
```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

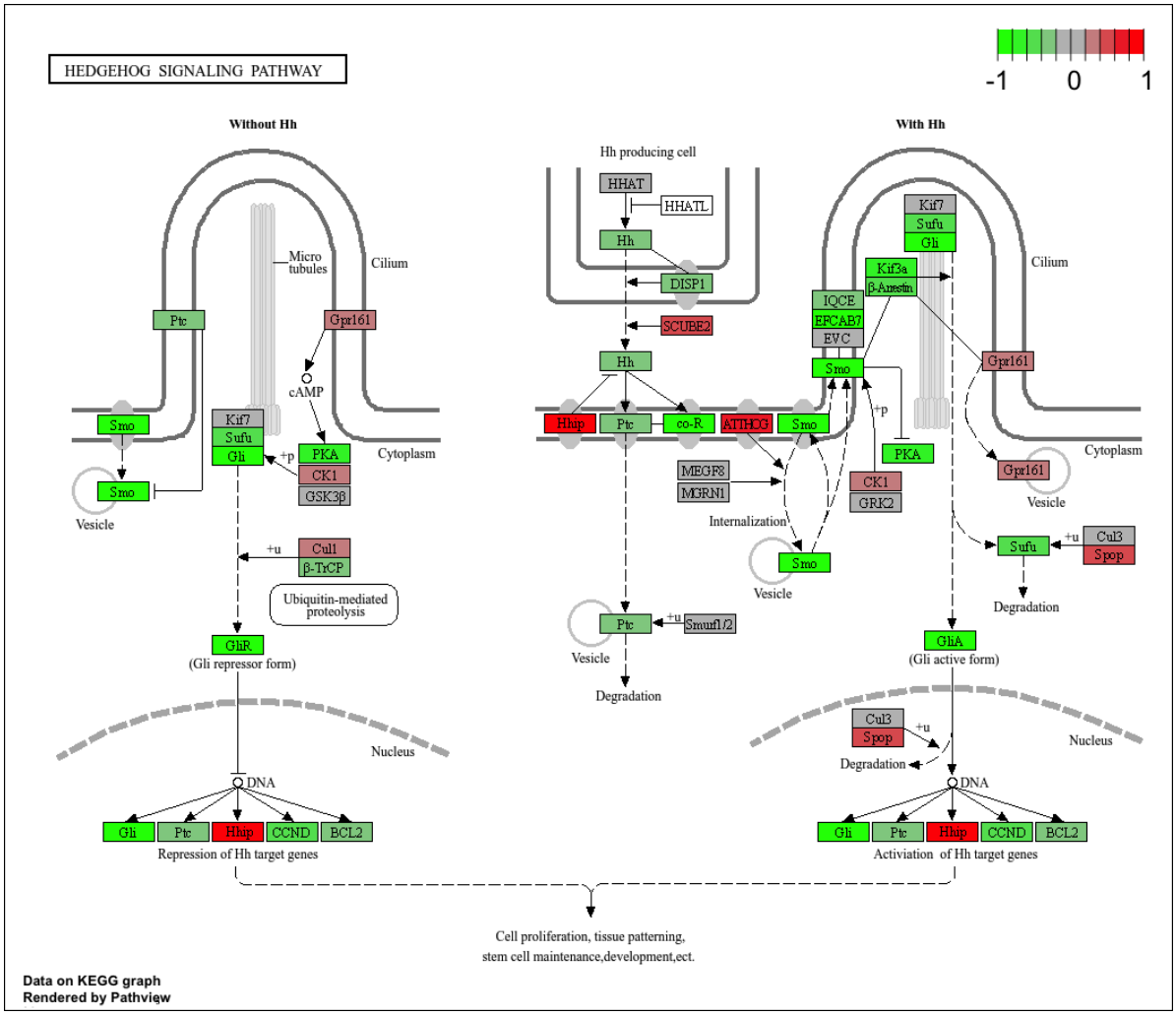
```
pathview(gene.data=foldchanges, pathway.id=keggresidsd, species="hsa")
```

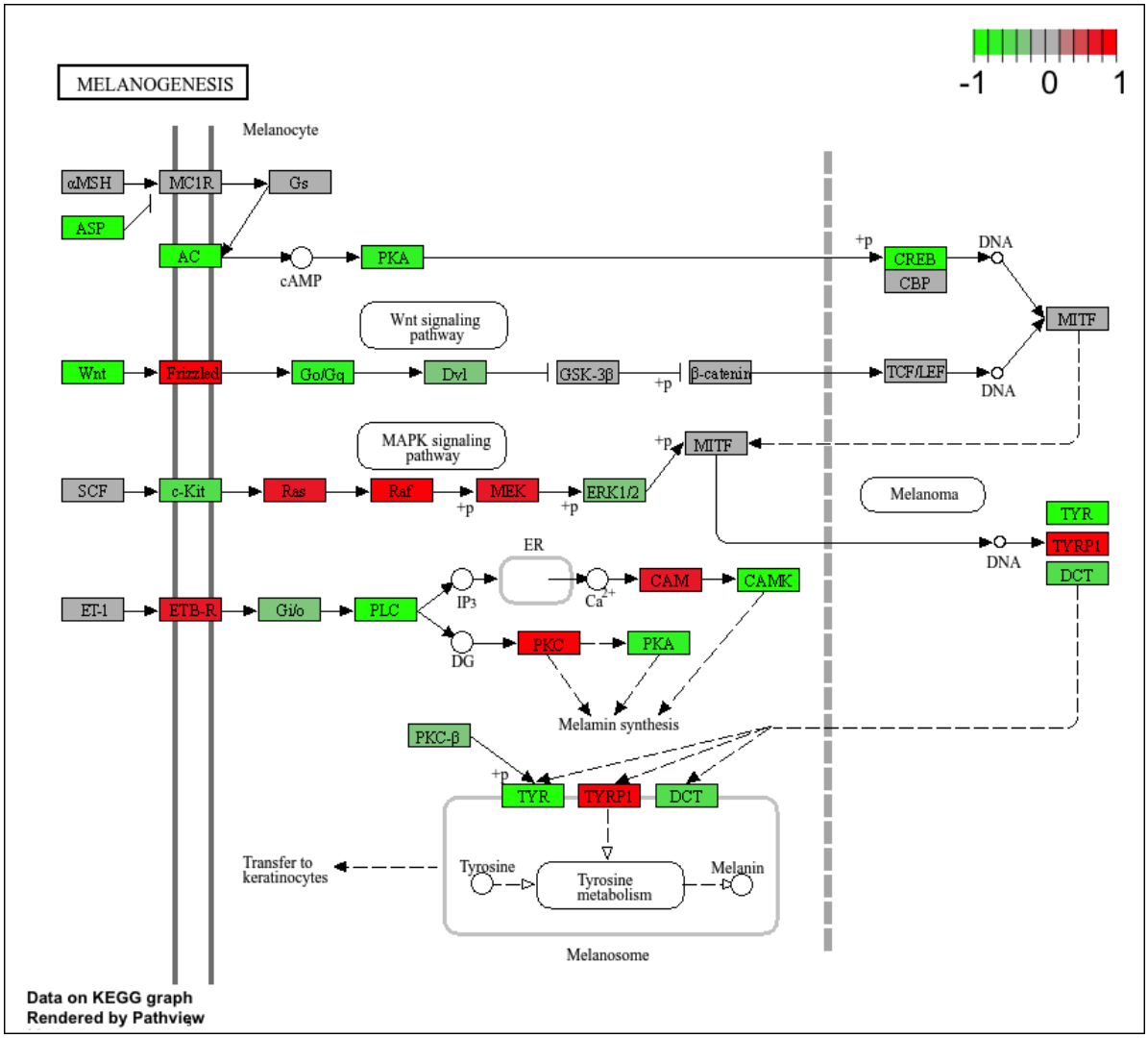
```
'select()' returned 1:1 mapping between keys and columns
```

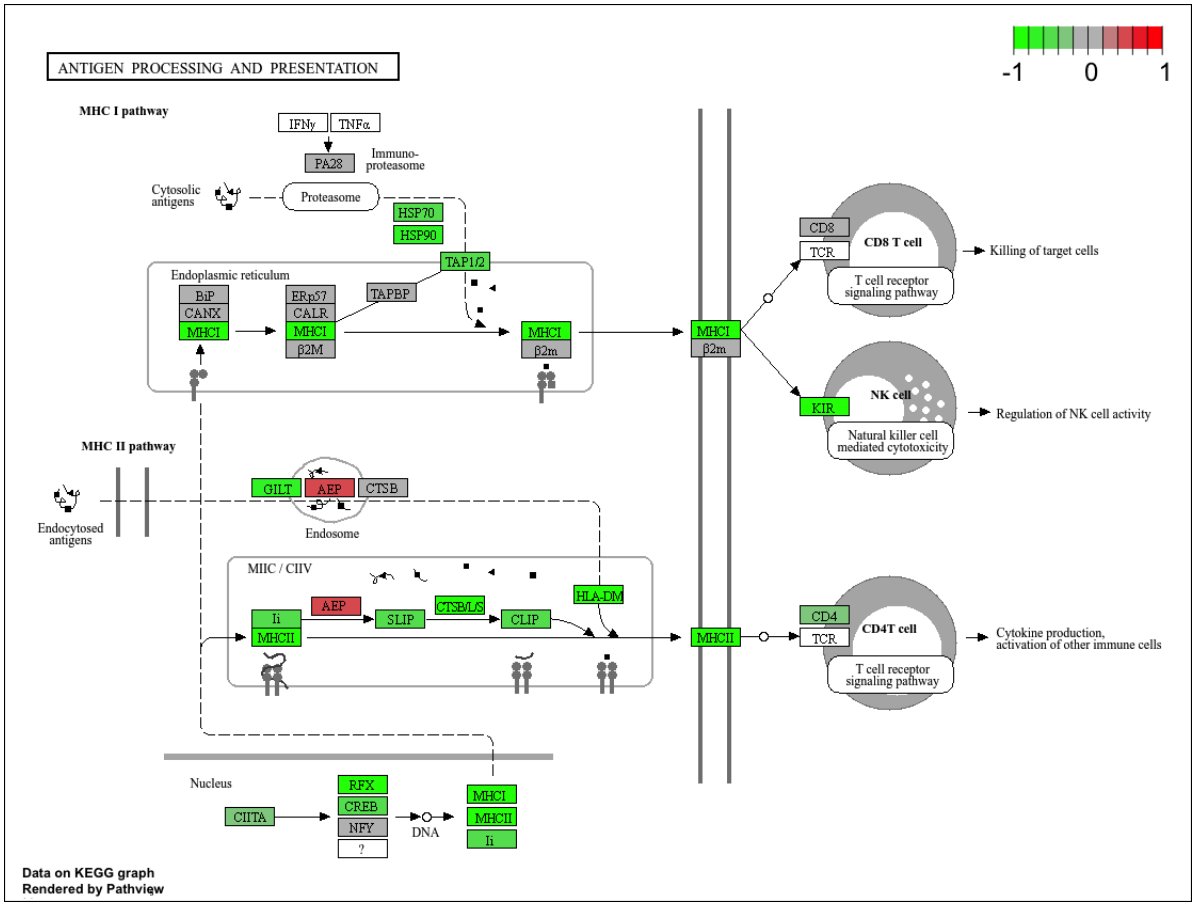
```
Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project
```

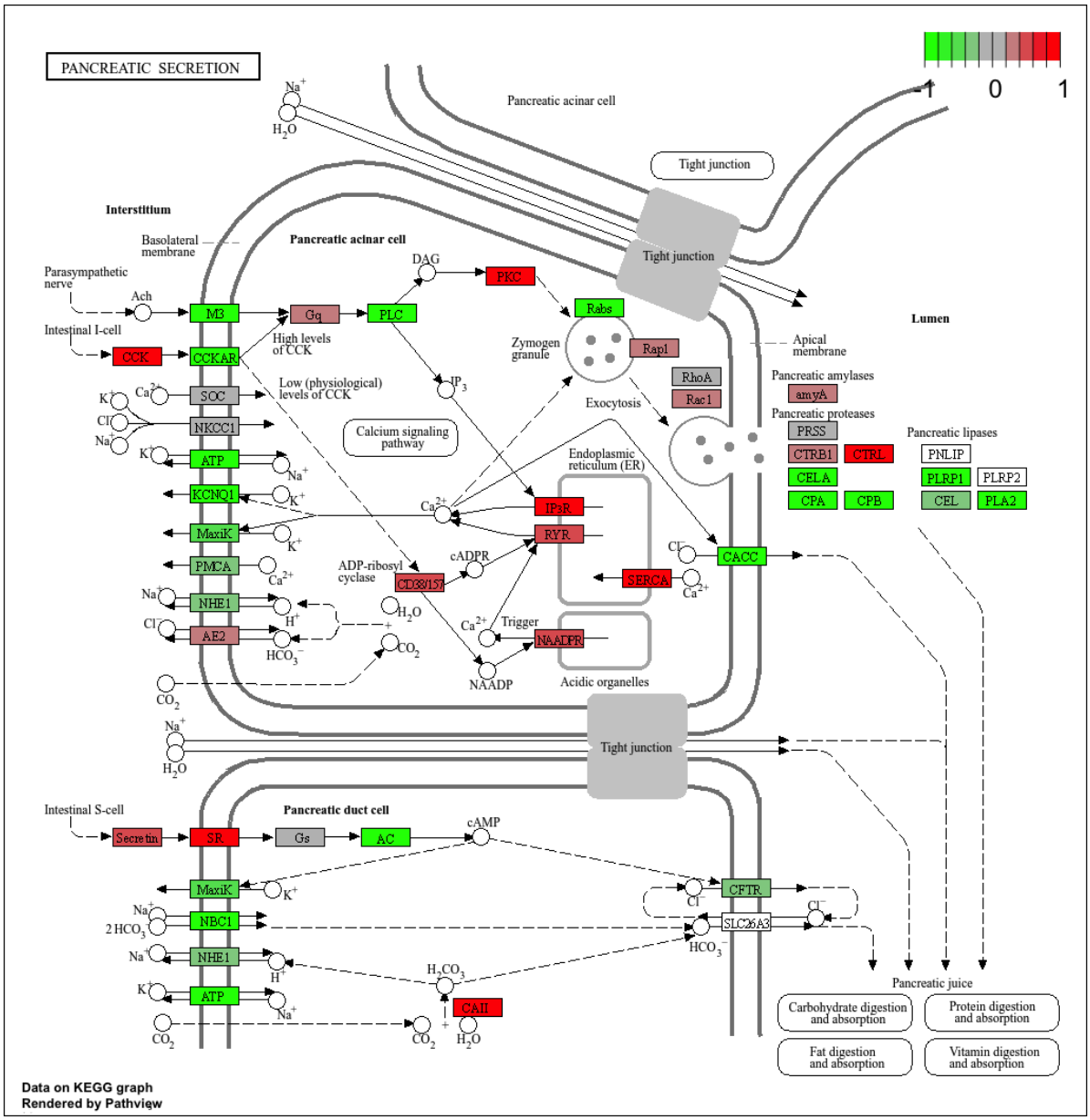
```
Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/paigecontreras/BIMM 143/class14: RNA-
Seq analysis mini-project

Info: Writing image file hsa04114.pathview.png
```


INTESTINAL IMMUNE NETWORK FOR IgA PRODUCTION
-1
0
1
Bacteria
Antigen
Antigen
M cell
Epitherial cell
SIgA
Gut lumen
Inductive site
Effector site
Lamina propria
Isolated lymphoid follicles
DC
IL-2
IL-4
IL-5
IL-6
IL-10
TGFβ
B7
MHC2
classII
APRIL
BAFF
BCR
TACI
BCMA
BAFFR
AID
RA
CCR9
α4β7
IgA+ B cell
B1 cell
Antigen processing and presentation
CD28
TCR
CD40L
ICOS
T cell receptor signaling pathway
CD4+ T cell
B2 cell
B cell proliferation
Class switch recombination
Somatic hypermutation
Peyer's patch
B cell receptor signaling pathway
Vascular system
High endothelial venules
CXCR4
CCR10
CCR9
α4β7
MADCAM
CXCL12
CCL28
CCL25
IgA+ plasmablast
Recirculaion
Leukocyte transendothelial migration
Homing
Terminal B cell differentiation
IgA
IgA-plasma cell
Lamina propria stroma cell
IL-15R
IL-15
LTβR
NFκ
Data on KEGG graph
Rendered by Pathview


HEDGEHOG SIGNALING PATHWAY
-1
0
1
Without Hh
With Hh
Hh producing cell
HHAT
HHATL
Hh
DISP1
SCUBE2
Micro tubules
Cilium
Ptc
Gpr161
cAMP
Smo
Kif7
Sufu
Gli
+p
PKA
CK1
GSK3β
Cytoplasm
Vesicle
+u
Cul1
β-TrCP
Ubiquitin-mediated proteolysis
GliR
(Gli repressor form)
Nucleus
DNA
Gli
Ptc
Hhip
CCND
BCL2
Repression of Hh target genes
Hhip
co-R
ATTHOG
MEGF8
MGRN1
Internalization
Smurf1/2
Degradation
IQCE
EFCAB7
EVC
Kif3a
β-Arrestin
GRK2
Cul3
Spop
GliA
(Gli active form)
Activiation of Hh target genes
Cell proliferation, tissue patterning,
stem cell maintenance,development,ect.
Data on KEGG graph
Rendered by Pathview


MELANOGENESIS
-1
0
1
Melanocyte
αMSH
MC1R
Gs
ASP
AC
cAMP
PKA
+p
CREB
CBP
DNA
MITF
Wnt signaling pathway
Wnt
Frizzled
Go/Gq
Dvl
GSK-3β
+p
β-catenin
TCF/LEF
DNA
MAPK signaling pathway
+p
MITF
SCF
c-Kit
Ras
Raf
MEK
ERK1/2
+p
+p
Melanoma
TYR
TYRP1
DCT
DNA
ER
ET-1
ETB-R
Gi/o
PLC
IP3
Ca2+
CAM
CAMK
DG
PKC
PKA
Melamin synthesis
PKC-β
+p
TYR
TYRP1
DCT
Transfer to keratinocytes
Tyrosine
Tyrosine metabolism
Melanin
Melanosome
Data on KEGG graph
Rendered by Pathview


ANTIGEN PROCESSING AND PRESENTATION
-1
0
1
MHC I pathway
IFNγ
TNFα
PA28
Immuno-proteasome
Cytosolic antigens
Proteasome
HSP70
HSP90
TAP1/2
Endoplasmic reticulum
BiP
CANX
MHCI
ERp57
CALR
MHCI
β2M
TAPBP
MHCI
β2m
MHCI
β2m
CD8
TCR
CD8 T cell
T cell receptor signaling pathway
Killing of target cells
KIR
NK cell
Natural killer cell mediated cytotoxicity
Regulation of NK cell activity
MHC II pathway
Endocytosed antigens
GILT
AEP
CTSB
Endosome
MIIC / CIIV
Ii
MHCII
AEP
SLIP
CTSB/LS
CLIP
HLA-DM
MHCII
CD4
TCR
CD4T cell
T cell receptor signaling pathway
Cytokine production, activation of other immune cells
Nucleus
CIITA
RFX
CREB
NFY
?
DNA
MHCI
MHCII
Ii
Data on KEGG graph
Rendered by Pathview




```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)
```

```r
lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.519724e-05  3.824205 8.519724e-05
GO:0002009 morphogenesis of an epithelium 1.396681e-04  3.653886 1.396681e-04
GO:0048729 tissue morphogenesis           1.432451e-04  3.643242 1.432451e-04
GO:0007610 behavior                       1.925222e-04  3.565432 1.925222e-04
GO:0060562 epithelial tube morphogenesis  5.932837e-04  3.261376 5.932837e-04
GO:0035295 tube development               5.953254e-04  3.253665 5.953254e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1951953      113 8.519724e-05
GO:0002009 morphogenesis of an epithelium 0.1951953      339 1.396681e-04
GO:0048729 tissue morphogenesis           0.1951953      424 1.432451e-04
GO:0007610 behavior                       0.1967577      426 1.925222e-04
GO:0060562 epithelial tube morphogenesis  0.3565320      257 5.932837e-04
GO:0035295 tube development               0.3565320      391 5.953254e-04

$less
                                             p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.536227e-15 -8.063910 1.536227e-15
GO:0000280 nuclear division              4.286961e-15 -7.939217 4.286961e-15
GO:0007067 mitosis                       4.286961e-15 -7.939217 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.169934e-14 -7.797496 1.169934e-14
GO:0007059 chromosome segregation        2.028624e-11 -6.878340 2.028624e-11
GO:0000236 mitotic prometaphase          1.729553e-10 -6.695966 1.729553e-10
                                                q.val set.size         exp1
GO:0048285 organelle fission             5.841698e-12      376 1.536227e-15
GO:0000280 nuclear division              5.841698e-12      352 4.286961e-15
GO:0007067 mitosis                       5.841698e-12      352 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.195672e-11      362 1.169934e-14
GO:0007059 chromosome segregation        1.658603e-08      142 2.028624e-11
GO:0000236 mitotic prometaphase          1.178402e-07       84 1.729553e-10

$stats
                                           stat.mean     exp1
GO:0007156 homophilic cell adhesion         3.824205 3.824205
GO:0002009 morphogenesis of an epithelium   3.653886 3.653886
GO:0048729 tissue morphogenesis             3.643242 3.643242
GO:0007610 behavior                         3.565432 3.565432
GO:0060562 epithelial tube morphogenesis    3.261376 3.261376
GO:0035295 tube development                 3.253665 3.253665
```

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

> Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The intestinal immune network has the most significant pathway, with a p-value of. 0.01006194. The pathways do not match in KEGG results, when ran again, instead the msot significant is hsa00500 Starch and sucrose metabolism with a p value of 0.007619766. Factors that can cause differences between the two methods include, KEGG pathways varying when being used, and GAGE outputting varying results.